

Collapse by Rotation: Training-Dynamics Neural Collapse in Variational Quantum Circuits, Verified on Real Vision Data

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Abstract

Papayan, Han, and Donoho (2020) established Neural Collapse (NC) as a terminal-phase geometric regularity of classical deep classifiers; Du, Yang, Tao, and Hsieh (2023) proved a quantum analogue holds at the *static global optimizer* of a regularized quantum classifier. Whether gradient-based training of a variational quantum circuit (VQC) ever reaches that optimum, and by what geometric route, has remained open. We show that a trained VQC collapses only in the $\mathcal{C}$ -dimensional measurement-visible subspace read by the loss, while, for a broad and practically relevant architecture class, the full representation’s Hilbert–Schmidt geometry is provably and numerically ($\sim 10^{-15}$ relative change) invariant under training: collapse is a training-driven rotation, not a contraction. We verify this mechanism to machine precision on a controlled synthetic benchmark and, newly, on real vision data (MNIST, Fashion-MNIST) under two independent encodings, replicate it at three qubit counts ($n = 4, 6, 8$), and validate it, within a strictly limited inference-only scope, on two independent IBM hardware backends. We further contrast the quantum representation’s exactly conserved total variance against a matched classical multi-layer perceptron, whose analogous within-class scatter grows by one to two orders of magnitude under training even as its normalized collapse ratio falls – a distinction invisible to normalized diagnostics alone. Every preregistered prediction in this campaign is reported against its scored outcome, confirmed, refuted, and partial alike, with an explicit accounting of the campaign’s full item count.

Keywords: neural collapse, variational quantum circuits, representation geometry, measurement subspace, terminal phase of training

1 Introduction

Neural Collapse (NC) is the terminal-phase phenomenon in which a deep classifier’s within-class features shrink toward their class means while the class means themselves converge to a simplex equiangular tight frame (ETF) (Papayan et al., 2020). Du et al. (2023) proved a quantum analogue: at the global optimizer of a regularized quantum classifier’s risk, feature states concentrate around per-class means and class-mean measurement operators become mutually orthogonal. That result characterizes a *static optimum*; it says nothing about whether, when, or how a variational quantum circuit (VQC) trained by gradient descent actually reaches it. This paper studies that gap directly: we track collapse metrics epoch by epoch, we use density-matrix trace distance and Hilbert–Schmidt geometry (operationally meaningful, quantum-information quantities) as our primary tools rather than working only in measurement-operator space, and we test entanglement’s causal role, encoding choice, ansatz family, optimizer, qubit count, and inference-time noise through controlled ablations and replications.

Our first finding is a null: every full-Hilbert-space collapse witness we compute – unnormalized and normalized trace distance, a Fisher-style ratio, purity, ETF deviation – remains statistically indistinguishable from an untrained random-initialization baseline, both before and after an overparameterization transition that otherwise governs whether the circuit interpolates the training set at all. We do not treat this null as a dead end. Cross-entropy, the loss that trains these circuits, depends on the state only through a C -dimensional measurement projection $z = (\langle Z_0 \rangle, \dots, \langle Z_{C-1} \rangle)$, where C is the number of classes. Full-space witnesses sum variance over all $4^n - 1$ traceless Hermitian directions of an n -qubit system and are dominated by directions the loss cannot see. Restricting attention to the z -projection recovers a collapse signal that is both real and large.

We then show, for a broad architecture class (fixed encoding applied exactly once, fixed measurement), that this restricted collapse is not a genuine contraction of the state constellation at all: it is a rigid rotation, and we verify this to machine precision on three independent quantities – total within-class Hilbert–Schmidt variance, per-class purity, and pairwise class-mean overlap. The dichotomy replicates at three qubit counts ($n = 4, 6, 8$), on real vision data (MNIST, Fashion-MNIST) under two independent encoding families (angle embedding, amplitude embedding), against a matched classical multi-layer perceptron baseline, across a second optimizer and a second ansatz family, and, within its licensed scope, on two independent quantum hardware backends.

This paper’s contributions:

1. A formal rotation theorem (Proposition 3) showing that, for fixed encoding and fixed measurement, training acts as a single global unitary conjugation on each sample’s pre-image state – an exact isometry of the full within-class Hilbert–Schmidt geometry – with structural corollaries for per-class purity and pairwise class-mean

overlap, and a consequence for the reachability of [Du et al.](#)’s optimum under this architecture.

2. Verification of the rotation mechanism at machine precision ($\sim 10^{-15}$ relative change) on a controlled synthetic benchmark, at three qubit counts, under a second optimizer and a second ansatz family, and, newly, on real vision data (MNIST, Fashion-MNIST) under both angle and amplitude encoding, with the terminal phase of training reached for the first time on real image features in this line of work once the per-class sample budget is reduced.
3. A representation-geometry contrast against a matched classical multi-layer perceptron, independently on two datasets: the quantum representation’s absolute scale is conserved under an exact symmetry, while the classical network’s analogous within-class scatter *grows* by one to two orders of magnitude under training even as its normalized collapse ratio falls – a distinction invisible to normalized NC diagnostics alone.
4. A dimension-counting account (Proposition 6) of why full-space collapse diagnostics have repeatedly failed to detect collapse in prior quantum NC work, replicated at three qubit counts, and a methodological caution (Remark 2) about a rank-deficiency instability in the standard NC1 estimator at small class counts, encountered directly in our own hardware analysis.
5. An inference-only hardware validation on two independent IBM backends (`ibm_marrakesh`, `ibm_kingston`), and a full accounting of every preregistered prediction made across this project’s ten result-bearing experimental stages, with an explicit statement of the counting convention used and a resolution of an internal discrepancy between two previously reported item-count figures.

Section 2 places this work against four related threads. Section 3 fixes model, loss, witness definitions, and the AI-assistance protocol used to produce this paper. Section 4 states and proves the paper’s central theoretical claims. Sections 5–9 present results in the order we obtained them: the synthetic controlled arm, real vision-data verification, the classical-versus-quantum comparative study, generality across optimizer/ansatz/dimension/noise, and hardware validation. Section 10 states limitations. Appendix A reproduces every preregistered prediction made in this campaign and its scored outcome, with the counting convention stated explicitly; Appendix B gives additional figures and traceability notes.

2 Related Work

Classical Neural Collapse. [Papayan et al. \(2020\)](#) introduced NC as an empirical terminal-phase phenomenon; [Han et al. \(2022\)](#) studied its dynamics under MSE loss along a “central path.” The unconstrained-features model (UFM) literature ([Mixon et al., 2022](#); [Yang et al., 2022](#)) studies NC as an optimization-landscape property of an idealized, norm-unconstrained penultimate representation – relevant to our comparative study (Section 7), where we find a real classical network’s representation growing rather than saturating under weak (but nonzero) weight decay, in the same unconstrained-feature spirit but with an empirically opposite absolute-scale trend to what a naive contraction analogy would predict. Follow-up landscape analyses sharpen

the UFM picture considerably: [Zhu et al. \(2021\)](#) show that cross-entropy with weight decay under the UFM has a benign global landscape whose only minimizers are simplex ETFs, [Tirer and Bruna \(2022\)](#) extend this to a richer feature model under MSE loss, and [Fang et al. \(2021\)](#) identify a complementary imbalanced-data failure mode (Minority Collapse) via the same layer-peeled modeling approach; [Kothapalli \(2023\)](#) surveys this line of work. None of it addresses a quantum representation or a measurement-subspace projection, which is precisely the gap our full-space/ z -space dichotomy targets.

Representation-geometry comparison tools. A separate classical literature asks how to compare learned representations directly: [Raghu et al. \(2017\)](#) (SVCCA) and [Kornblith et al. \(2019\)](#) (centered kernel alignment) both develop transformation-invariant similarity measures for exactly this purpose. Our full-space-vs-measurement-subspace decomposition (Section 4) is a structural, not statistical, split of representation geometry specific to the quantum setting, but targets the same underlying question these tools were built to answer classically: what changes, and what provably cannot change, in a learned representation under training.

Quantum NC and expressivity. [Du et al. \(2023\)](#) is this paper’s primary point of departure (Section 1). [Larocca et al. \(2023\)](#) bound VQC overparameterization transitions via the dynamical Lie algebra (DLA); our capacity-transition results (Section 5) are consistent with, but sit strictly below, their DLA-derived upper bound, and a single-input Fisher-rank saturation point we identify is tighter than their bound for single-input measurements. [Bowles et al. \(2024\)](#) caution against overinterpreting entanglement’s necessity for generalization; our entanglement-ablation finding (a hard capacity ceiling for a non-entangling ansatz) is scoped strictly to exact training-set interpolation, not generalization, and the two findings are compatible. [San Sebastian et al. \(2026\)](#) study observable-set choices for multi-class quantum classification in measurement-operator space, complementary to our density-matrix-space and z -space treatment. [Mondal et al. \(2026\)](#) introduce a learnable measurement-temperature mechanism conceptually analogous to the learnable logit scale in our readout family, which we find plays a matching accelerant role.

Quantum feature encodings and training landscapes. Our VQC classifier’s data encoding is an instance of the quantum-kernel-methods view formalized by [Schuld and Killoran \(2019\)](#) and demonstrated experimentally by [Havlíček et al. \(2019\)](#), both of which treat the encoding step as a nonlinear feature map into a quantum Hilbert space – the same encoding step our rotation theorem (Proposition 3) analyzes structurally rather than as a kernel. [Abbas et al. \(2021\)](#) propose an effective-dimension measure of quantum model capacity and trainability, relevant background for our capacity-transition results (Section 5); [Cerezo et al. \(2021\)](#) survey variational quantum algorithms broadly. [McClean et al. \(2018\)](#) identify barren plateaus as a trainability obstruction that grows with circuit width for random circuits; our capacity-transition and TPT-reachability results describe a regime that trains successfully, consistent with staying away from that pathology rather than a direct study of it.

3 Model and Methods

3.1 Model

We consider an n -qubit VQC classifier over C classes. Each input x is encoded via a fixed-form angle embedding (R_Y rotations parameterized by x , PCA-reduced, standardized, and scaled to $[-\pi, \pi]$ per feature) or, in the amplitude-encoding variant, via `AmplitudeEmbedding( $x$ )`, into a pre-image pure state $\rho_0(x) = |\psi_0(x)\rangle\langle\psi_0(x)|$. A parameterized ansatz $U(\theta)$ is applied – PennyLane’s `StronglyEntanglingLayers` with L layers by default, or, as a generality check, a hardware-efficient alternative (per-qubit R_Y - R_Z rotations with a linear-chain CNOT entangler) – and the model reads out $z(\theta; x) = (\langle Z_0 \rangle, \dots, \langle Z_{C-1} \rangle) \in \mathbb{R}^C$ from the first C qubits. Logits are $\beta \cdot z$ with a learnable inverse-temperature (scale) β , trained by cross-entropy. All simulator runs use exact statevector simulation (`default.qubit`, no shot noise) unless stated otherwise (Section 8’s noisy-inference arm uses `default.mixed`). We say training reaches the *terminal phase of training* (TPT) when training accuracy saturates at 1.0 for a sustained window of epochs; runs that never reach TPT are excluded from collapse-magnitude conclusions but retained for full-space invariance conclusions (Section 6).

Two architectural switches matter throughout this paper:

- **Data re-uploading.** When `reupload=false`, the encoding is applied exactly once, so $\rho_0(x)$ is θ -independent. When `reupload=true`, the encoding is re-applied at intermediate layers under the current trained weights, so the map from x to the pre-measurement state is itself trained.
- **Measurement trainability.** The readout family R_0 uses a fixed Pauli-Z measurement; family R_3 additionally trains a measurement-basis unitary.

The primary synthetic dataset, `blobs3`, is three Gaussian blobs in $\mathbb{R}^4$ (or $\mathbb{R}^6, \mathbb{R}^8$ for our qubit-count replications), chosen so that interpolation difficulty is controlled and the study measures optimization and geometry, not data pathology. Real-data verification (Section 6) uses MNIST and Fashion-MNIST, classes $\{0, 1, 2\}$, PCA-reduced to 4 (angle encoding) or 16 (amplitude encoding) dimensions. A capacity-matched classical multi-layer perceptron (MLP), trained on the identical PCA-4 features with weak L2 regularization (`wd` = 5×10^{-4}), serves as the comparative-study baseline (Section 7).

3.2 Full-space and measurement-subspace witnesses

Let $B = \text{span}\{B_0, \dots, B_{C-1}\}$ be the orthonormalized Pauli-Z generator span, a C -dimensional subspace of the $(4^n - 1)$ -dimensional space of traceless Hermitian operators. For a class-conditional constellation of density operators $\{\rho(x_i; \theta)\}$, define the total within-class Hilbert–Schmidt variance

$$\text{total_var}(\theta) = \sum_c \sum_{i \in c} \|\rho(x_i; \theta) - \bar{\rho}_c(\theta)\|_{HS}^2, \quad (1)$$

which decomposes, by Parseval’s identity over an orthonormal basis of traceless-Hermitian directions, as

$$\text{total_var} = V_{\text{meas}} + V_{\text{fiber}}, \quad (2)$$

where V_{meas} is the part of the variance lying in B (visible to the loss) and V_{fiber} is the complementary part (the “fiber,” invisible to the loss); $\text{fiber_fraction} = V_{\text{fiber}}/(V_{\text{meas}} + V_{\text{fiber}})$. Full-space witnesses also include within-class trace distance, per-class purity $\text{tr}(\bar{\rho}_c^2)$, and deviation from the simplex-ETF angular target, all computed over the full $4^n - 1$ -dimensional space.

The z -space analogue of classical NC1,

$$\text{NC1}_z = \frac{1}{C} \text{tr}(\Sigma_W^z \text{pinv}(\Sigma_B^z)), \quad (3)$$

measures collapse restricted to the measured projection z , with Σ_W^z, Σ_B^z the within- and between-class scatter matrices of $z(\theta; x)$, and the scale-free ratio $r_z = \text{tr}(\Sigma_W^z)/\text{tr}(\Sigma_B^z)$ (Remark 2 motivates reporting both). A box-vertex geometry model and a simplex-ETF geometry model are compared directly on the centered class means (Section 4.1).

3.3 Preregistration and scoring

Every prediction reported in this paper was written and committed to a preregistration record before the corresponding metric was computed, and is scored strictly as CONFIRMED, REFUTED, PARTIAL, or, where no directional threshold applies (e.g. a scoping probe), reported factually and marked UNSCOREABLE. Refuted and partial results are reported plainly, not omitted or softened. This paper’s campaign spans ten result-bearing experimental stages; Appendix A reproduces every preregistered item and its scored outcome, with the counting convention used to total them stated explicitly.

3.4 AI-assistance disclosure

The implementation and drafting of this paper were carried out with the assistance of an AI coding agent (a large-language-model-based coding assistant), operating under a frozen mathematical specification written by the author before implementation began, preregistered predictions committed to a version-controlled record before each corresponding result was computed (Section 3.3), and an automated test suite (173 unit and integration tests at the time of writing, covering every function in the metrics-computation module against the specification’s formulas) that was required to pass, unmodified in its assertions, before and after every implementation change. The AI agent’s role spanned experiment orchestration, metrics-code implementation against the frozen specification, and drafting of manuscript prose from the resulting numerical records. The author designed the specification, the preregistered predictions and their thresholds, and the overall research questions; reviewed every reported number against its source log file; and is responsible for all content in this paper, having independently verified the theorem statements and proofs in Section 4. This disclosure follows the journal’s policy that use of large language models in research and writing be documented in the Methods section rather than treated as unlisted authorship; no AI tool is credited as an author.

Collapse by rotation

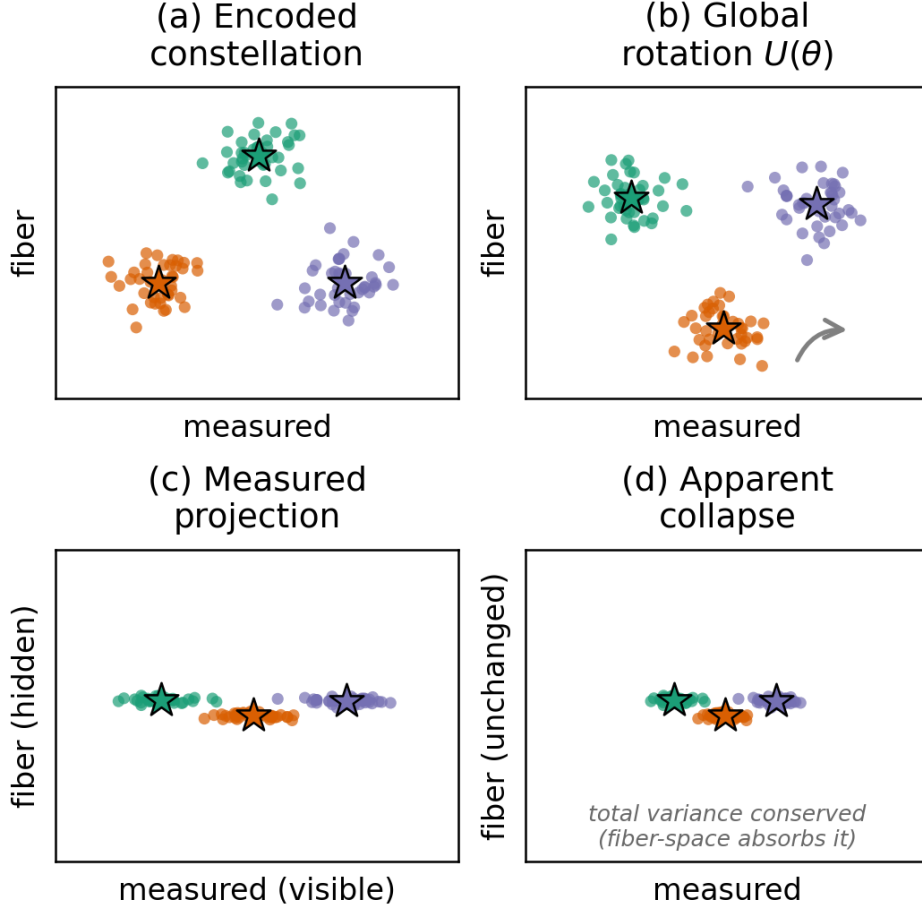


Fig. 1 Schematic of the measurement-subspace/fiber decomposition and the rotation mechanism: the C -dimensional measured subspace B receives all of the loss gradient (Lemma 1); under fixed encoding and fixed measurement, training rotates the full state constellation rigidly in Hilbert space (Proposition 3), redistributing a conserved variance budget between V_{meas} and V_{fiber} .

4 Theory: Collapse Is a Rotation

Lemma 1 (Loss factors through the measured subspace). *For the readout of Section 3.1, the cross-entropy loss $L(\theta)$ depends on θ only through $z(\theta) = (\langle Z_0 \rangle, \dots, \langle Z_{C-1} \rangle)$. Consequently, for any perturbation direction H in the tangent space of density operators at $\rho(x; \theta)$ that is Hilbert–Schmidt-orthogonal to B for every sample, the loss gradient in that direction vanishes:*

$$\left. \frac{dL}{d\theta} \right|_{\text{restricted to } \langle H, B_j \rangle_{HS} = 0 \ \forall j} = 0.$$

Proof. L is a composition of the softmax-cross-entropy map with $\theta \mapsto \beta z(\theta)$; by the chain rule its gradient lies entirely in the span of $\partial z_j / \partial \theta$ for $j = 0, \dots, C-1$. Each $z_j = \text{tr}(B_j \rho(x; \theta))$ is itself a linear functional of ρ restricted to B_j ; its differential along any direction H with $\langle H, B_j \rangle_{HS} = 0$ for all j is $\text{tr}(B_j H) = 0$ by definition of the Hilbert–Schmidt inner product, for every j . Hence $dz_j/d\theta$ restricted to such H is 0 for all j , and by the chain rule $dL/d\theta$ restricted to H is 0. $\square$

Corollary 2 (No loss-selected pressure along fibers). *Gradient descent applies zero pressure, in either sign, along fiber directions (the Hilbert–Schmidt complement of B) at every training step, for any trajectory, regardless of encoding. Full-space collapse diagnostics summing variance over all $4^n - 1$ directions are therefore diluted by exactly this zero-gradient subspace whenever $C \ll 4^n - 1$ – the structural explanation for the full-space null of Section 5.*

Proposition 3 (Collapse by rotation). *Suppose the encoding is applied exactly once (`reupload=false`), so each sample’s pre-image state $\rho_0(x)$ is θ -independent, and the entire trained circuit acts as a single global unitary $U(\theta)$ applied after encoding:*

$$\rho(x; \theta) = U(\theta) \rho_0(x) U(\theta)^\dagger. \quad (4)$$

Then $\text{total_var}(\theta)$ (Eq. 1) is exactly invariant under training: $\text{total_var}(\theta) = \text{total_var}(\theta_0)$ for every θ reachable from initialization θ_0 by any gradient trajectory.

Proof. **Step 1 (linearity of conjugation over the class average).** For a fixed unitary $U(\theta)$, conjugation $A \mapsto U(\theta)AU(\theta)^\dagger$ is linear in A . Hence the class-mean operator commutes with taking the average:

$$\bar{\rho}_c(\theta) = \frac{1}{N_c} \sum_{i \in c} \rho(x_i; \theta) = \frac{1}{N_c} \sum_{i \in c} U(\theta) \rho_0(x_i) U(\theta)^\dagger = U(\theta) \bar{\rho}_c^{(0)} U(\theta)^\dagger,$$

where $\bar{\rho}_c^{(0)} = \frac{1}{N_c} \sum_{i \in c} \rho_0(x_i)$ is the θ -independent pre-image class mean.

Step 2 (cyclic-trace isometry of the Hilbert–Schmidt norm). For any operator D and unitary U ,

$$\|UDU^\dagger\|_{HS}^2 = \text{tr}(UDU^\dagger(UDU^\dagger)^\dagger) = \text{tr}(UDD^\dagger U^\dagger) = \text{tr}(DD^\dagger) = \|D\|_{HS}^2,$$

using unitarity ($U^\dagger U = I$) and cyclicity of trace. Applying this with $D = \rho_0(x_i) - \bar{\rho}_c^{(0)}$ (using Step 1) gives, for every sample i in class c and every θ ,

$$\|\rho(x_i; \theta) - \bar{\rho}_c(\theta)\|_{HS}^2 = \|U(\theta)(\rho_0(x_i) - \bar{\rho}_c^{(0)})U(\theta)^\dagger\|_{HS}^2 = \|\rho_0(x_i) - \bar{\rho}_c^{(0)}\|_{HS}^2,$$

independent of θ .

Step 3 (sum over samples). Summing the θ -independent per-sample quantity from Step 2 over all $i \in c$ and all classes c reproduces exactly $\text{total_var}(\theta)$ (Eq. 1), which is therefore equal to the same sum evaluated at $\theta = \theta_0$ (indeed, for *any* value of θ , since the right-hand side of Step 2 never involves θ). Hence total_var is exactly invariant under training. $\blacksquare$ $\square$

Corollary 4 (Purity/overlap invariance; Du et al.’s optimum is unreachable). *Under the hypotheses of Proposition 3, per-class-mean purity $\text{tr}(\bar{\rho}_c(\theta)^2)$ and pairwise class-mean overlap $\text{tr}(\bar{\rho}_c(\theta)\bar{\rho}_{c'}(\theta))$ are each exactly invariant under training, for every class c (respectively pair c, c'). Consequently, whatever pairwise class-mean overlap the untrained model starts with at initialization is exactly the overlap it ends with: Du et al.’s orthogonal-means optimum (overlap $\rightarrow 0$) is training-unreachable under this architecture unless it already holds at initialization.*

Proof. Purity: $\text{tr}((U\rho U^\dagger)^2) = \text{tr}(U\rho^2 U^\dagger) = \text{tr}(\rho^2)$ by cyclicity of trace, applied to $\rho = \bar{\rho}_c(\theta)$ using Step 1 of Proposition 3’s proof. Pairwise overlap: identically, $\text{tr}(U\bar{\rho}_c^{(0)}U^\dagger U\bar{\rho}_{c'}^{(0)}U^\dagger) = \text{tr}(U\bar{\rho}_c^{(0)}\bar{\rho}_{c'}^{(0)}U^\dagger) = \text{tr}(\bar{\rho}_c^{(0)}\bar{\rho}_{c'}^{(0)})$, again by cyclicity, for every θ . Reachability follows since the overlap value is pinned to its θ_0 value for all θ . $\square$

Scope. Proposition 3 requires (i) the encoding be applied exactly once and be θ -independent, and (ii) the measurement be fixed, not trained. Data re-uploading breaks hypothesis (i) outright (the pre-image state itself becomes trained), making real contraction possible; a trainable measurement block (R_3) breaks (ii). Neither is covered by Proposition 3 as stated.

Remark 1 (Entanglement entropy is not covered). *Proposition 3 and its corollaries apply only to basis-independent functionals of the full state under a global unitary. Entanglement entropy at a fixed bipartition is the von Neumann entropy of a reduced state and is invariant only under local (factored) unitaries, not under a generic global entangling $U(\theta)$. Nothing in this section implies entanglement entropy is constant under training; we make no such claim.*

4.1 ETF geometry and dimension counting

Proposition 5 (ETF directions without box saturation). *Under a learnable logit scale β , the centered measured-subspace class means $\tilde{m}_c = \bar{z}_c - \bar{z}_G$ converge in direction to the simplex-ETF angular target (pairwise cosine $-1/(C-1)$), while their magnitude $\|\tilde{m}_c\|$ remains far below the box-vertex saturation radius $\sqrt{8/3}$ (at $C = 3$). At $C = 3$ the centered box vertices themselves already have pairwise cosine exactly $-1/(C-1)$, so angular convergence alone cannot discriminate an ETF-direction model from a box-vertex model; radius is the discriminating statistic.*

Proof. The cosine claim is a direct algebraic computation on the hand-built box vertices v_c (+1 in coordinate c , -1 elsewhere): for $C = 3$, $v_c \cdot v_{c'} = -1$ and $\|v_c - \bar{v}\|^2 = 8/3$ for $c \neq c'$, giving pairwise cosine $-1/2 = -1/(C-1)$, matching the simplex-ETF angular target exactly. Cross-entropy’s margin pressure on logits $\beta\tilde{m}_c \cdot u$ (for a unit probe direction u) is satisfied by growing the scalar β without bound, since β is unconstrained and appears linearly; growing $\|\tilde{m}_c\|$ itself is not required once directions are separated, because the loss is a function of $\beta\|\tilde{m}_c\|$ jointly, not of $\|\tilde{m}_c\|$ alone. A free scalar multiplier therefore absorbs the margin-pressure term that would otherwise push $\|\tilde{m}_c\|$ toward a saturating value, leaving $\|\tilde{m}_c\|$ to converge wherever gradient descent’s remaining (weaker) pressure leaves it – empirically, to a radius far short of $\sqrt{8/3}$ (Section 5.3). $\square$

Proposition 6 (Fiber dimension counting). *The measured subspace B has dimension exactly C ; the ambient space of traceless Hermitian operators on n qubits has dimension $4^n - 1$. Under an isotropic-null assumption (variance spread uniformly over all $4^n - 1$ directions, as at random initialization), the expected fiber fraction is*

$$\text{fiber_fraction}_{\text{null}}(n) = 1 - \frac{C}{4^n - 1}, \quad (5)$$

strictly increasing in n for fixed C .

Proof. Direct dimension count: the fiber subspace is the Hilbert–Schmidt orthogonal complement of B within the $(4^n - 1)$ -dimensional traceless-Hermitian space, hence has dimension $4^n - 1 - C$. Under the stated isotropic-null assumption, variance is distributed uniformly per dimension, so the expected fiber share is the fiber’s dimension fraction, Eq. 5. Monotonicity in n follows since $C/(4^n - 1) \rightarrow 0$ as n grows for fixed C . $\square$

Remark 2 (NC1_z /pseudoinverse instability at small C). *For any C -class problem, $\Sigma_B^z \in \mathbb{R}^{C \times C}$ is exactly $\text{rank} \leq C - 1$, since class-mean deviations from the global mean sum to zero. Consequently $\text{pinv}(\Sigma_B^z)$ (Eq. 3) is computed against a matrix with at least one exactly-zero eigenvalue; any perturbation (empirical noise, hardware residual) that introduces a small-but-nonzero eigenvalue in that exact null direction is treated by `pinv`’s default tolerance as invertible, and the trace-with-`pinv` formula can blow up by orders of magnitude. We recommend reporting the inversion-free ratio $r_z = \text{tr}(\Sigma_W^z)/\text{tr}(\Sigma_B^z)$ alongside NC1_z whenever C is small or Σ_B^z may be empirically near-degenerate; Section 9 encounters this instability directly.*

Proof. Standard perturbation theory for pseudoinverses near a rank change: the pseudoinverse of a rank-deficient matrix perturbed by a generic full-rank term is discontinuous in the perturbation magnitude precisely along the original null direction. Concretely, $\sum_c (\bar{z}_c - \bar{z}_G) = 0$ by construction of the global mean $\bar{z}_G$, so the C row/column deviations of Σ_B^z are linearly dependent, forcing $\text{rank}(\Sigma_B^z) \leq C - 1$. $\square$

5 Results: The Synthetic Controlled Arm

We first isolate the rotation mechanism from data-specific confounds using the controlled `blobs3` benchmark (Section 3.1), before extending to real vision data (Section 6) and a classical comparative study (Section 7).

5.1 The full-space null and the capacity transition

A classical MLP control reproduced classical NC cleanly before any quantum measurement was made, validating our instrumentation. A three-class quantum anchor at 6 qubits never reached TPT across 17 tuning attempts, while classical baselines on identical features reached full accuracy; a two-class diagnostic that did interpolate showed post-TPT trace distance statistically indistinguishable from a 10-seed untrained null. A 40-run sweep over circuit depth $L \in \{2, 4, 8, 12, 16, 21, 26, 32\}$ (4 qubits, 3 classes, 5 seeds each) shows TPT reachability transitioning sharply with parameter count M :

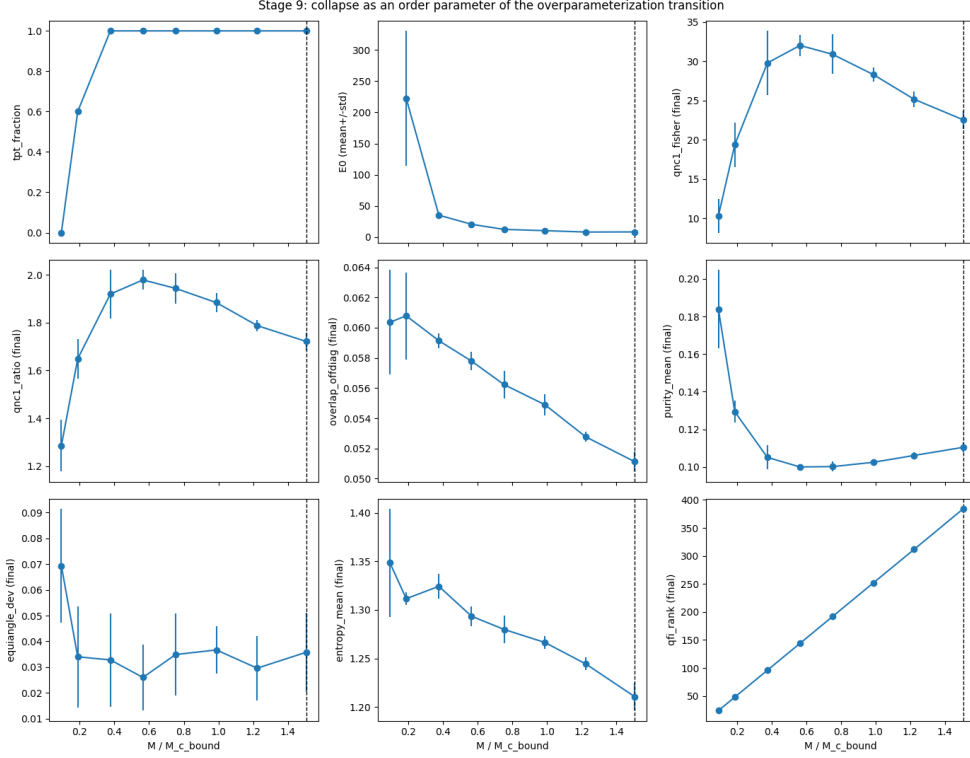


Fig. 2 TPT fraction, terminal epoch, and full-space collapse witnesses vs. the capacity ratio M/M_c^{bound} , 8 depths $\times$ 5 seeds. Collapse witnesses do not jump at the capacity transition.

0/5 seeds at $M/M_c^{\text{bound}} = 0.094$, 3/5 at 0.188, 5/5 from 0.376 onward, consistent with Larocca et al.’s overparametrization theory (Larocca et al., 2023). Figure 2 shows every outcome variable against this shared capacity axis. Even at full interpolation, every full-space collapse witness remains statistically indistinguishable from the untrained null (Fisher ratio $p = 0.75$, ETF deviation $p = 0.96$, underparameterized vs. overparameterized groups) – the structural consequence of Corollary 2. Removing entangling gates caps the ansatz’s dynamical-Lie-algebra dimension at 12 regardless of depth: the non-entangling ansatz reaches TPT in 0/3 seeds even at a parameter count where the entangling ansatz reaches 5/5.

5.2 Measurement-subspace collapse

Cross-entropy depends on the state only through z (Lemma 1); we predicted, and pre-registered before computing it, that collapse concentrates there while fiber directions do not. At the 4-qubit, $L=8$ anchor, NC1_z decays by a factor of 0.0045 from epoch 0 to the final epoch (5 seeds), and the scale-free margin $\beta \cdot \overline{\text{margin}}$ grows from -0.87 to 8.55. An untrained 10-seed null shows no such decay (Mann–Whitney $p = 6.7 \times 10^{-4}$). The fiber fraction reaches 0.9934, exceeding the preregistered 0.9 threshold, though it

stage 13 supplement: V_{fiber} dip vs untrained-null band (mean $\pm$ 1 std, $n=5$ seeds)

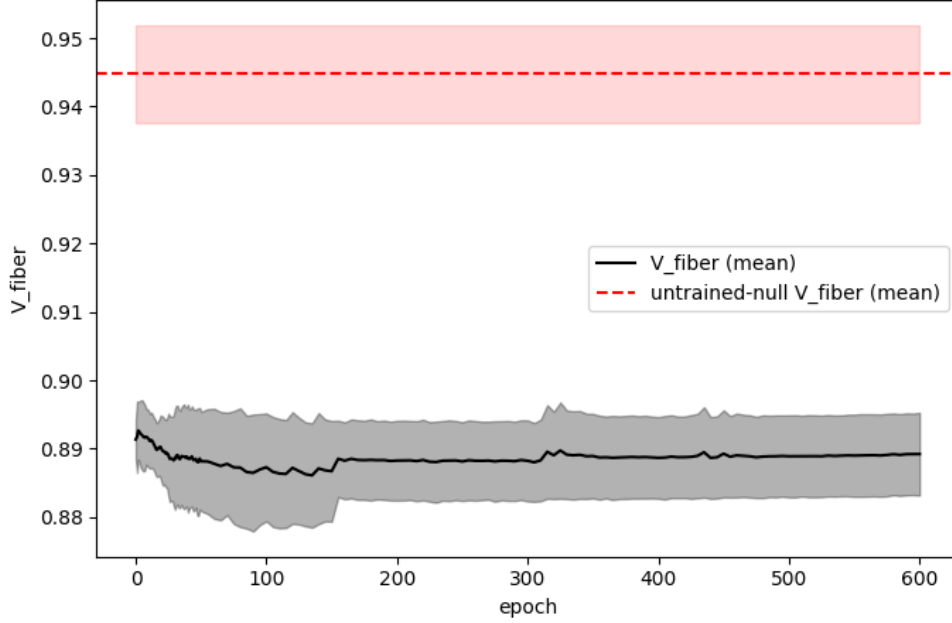


Fig. 3 Fiber-subspace within-class variance V_{fiber} vs. epoch, 4-qubit $L=8$ anchor, against the untrained-null band.

is not perfectly flat against the untrained-null band. Figure 3 shows the fiber-variance trajectory directly.

5.3 ETF directions without box saturation

At the 3-class anchor, the ETF-deviation witness reaches 0.0722 against a normalized box-vertex deviation of 0.7122. Per Proposition 5, angle alone cannot discriminate the two geometric models at $C=3$; recomputing the class-mean radius trajectory at the same anchor, the mean radius grows from 0.0649 ± 0.0164 at epoch 0 to 0.3287 ± 0.0257 at the final epoch – about 20% of the box-saturation value $\sqrt{8/3} \approx 1.633$. Directions converge to the shared angular target while magnitude stays far from saturation, consistent with β absorbing the scale that cross-entropy’s margin pressure would otherwise push into the state itself: raising β ’s ceiling from ~ 30 to 500 changes nothing bit-for-bit, while freezing β delays TPT onset by roughly $4.1\times$ (mean E_0 145.2 vs. 35.2 epochs) without forcing additional within-class collapse. Mondal et al. (2026) introduce an analogous learnable measurement-temperature parameter to improve gradient sensitivity in hybrid quantum classifiers; our frozen- β delay is an independent empirical echo of the same “temperature as accelerant” mechanism. Figure 4 shows the angular trajectory directly.

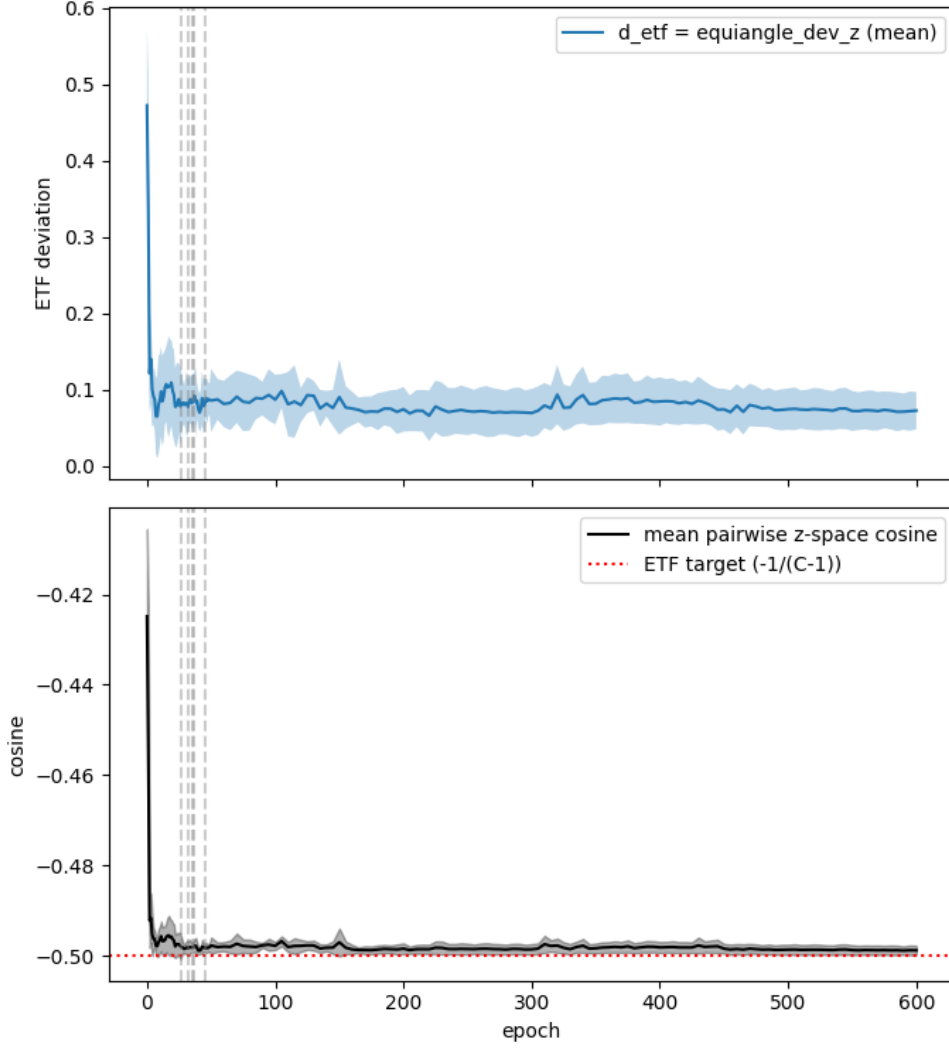


Fig. 4 ETF-deviation and pairwise z -space cosines vs. epoch, 4-qubit $L=8$ anchor, TPT onset marked per seed.

5.4 Measurement expressivity

We tested whether full-space collapse strengthens as the measurement becomes more expressive, across four readout families at matched parameter count. The apparent monotonic weakening we first observed across families was substantially a circuit-depth confound; fixing depth at $L=8$ removes the gap for every full-space witness tested ($p \geq 0.42$). The fiber-fraction gap that remains is explained by an isotropic

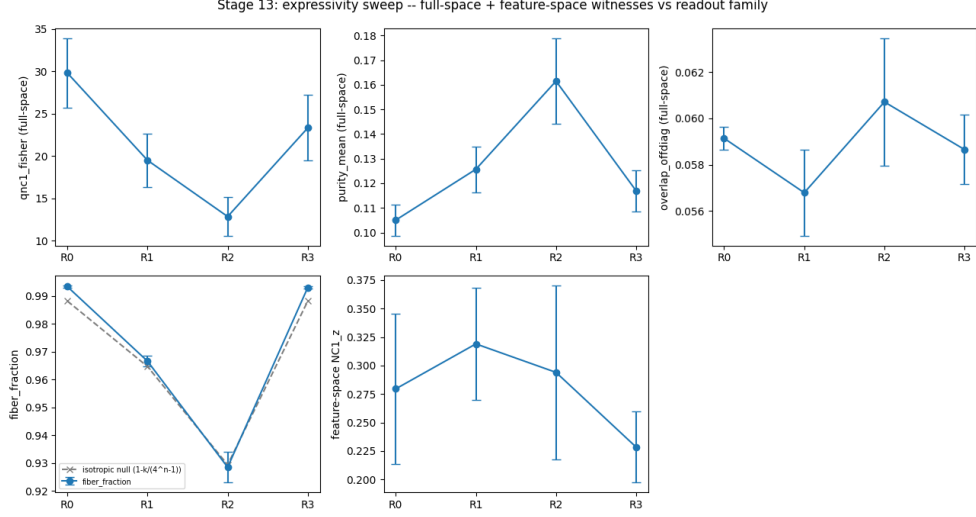


Fig. 5 Fiber fraction vs. readout family, matched and fixed-depth sweeps, isotropic dimension-counting null overlaid.

dimension-counting null (Proposition 6, generalized to $C \rightarrow k$) rather than by a strong measurement-subspace effect: loss-visible directions are only $1.4\text{--}1.7\times$ more collapsed than the measured-span average, short of the $2\times$ threshold we preregistered. A logit-space NC1 witness is statistically indistinguishable across all four families and both depths tested ($p \geq 0.31$ throughout). Figure 5 overlays the isotropic null against the observed fiber fractions.

5.5 Robustness of the within-class-variance floor

We tested six candidate mechanisms for why within-class z -space variance stays roughly flat rather than shrinking, while between-class separation grows. NC1_z decay itself is robust across every arm we tried: raising the logit-scale ceiling to 500 (uninformative, since the ceiling was never approached), a 5000-epoch horizon showing no slow drift (Spearman $\rho = -0.072$, $p = 0.48$), label smoothing (right direction, not significant at 5 seeds), and localizing the capacity-transition onset across three depths (TPT fraction $0.3 \rightarrow 1.0 \rightarrow 1.0$, non-decreasing as preregistered). Two arms directly refute candidate causal mechanisms: a single-input quantum Fisher information rank saturates at exactly $30 = 2(2^4 - 1)$ for every depth tested – the real dimension of the 4-qubit pure-state manifold rather than the parameter-space bound of 255 the batch-aggregated rank approaches; and freezing the logit scale does *not* force additional within-class collapse – if anything, variance shrinks less, in the wrong direction for the “escape valve” hypothesis we had proposed. Figure 6 shows the long-horizon stability directly.

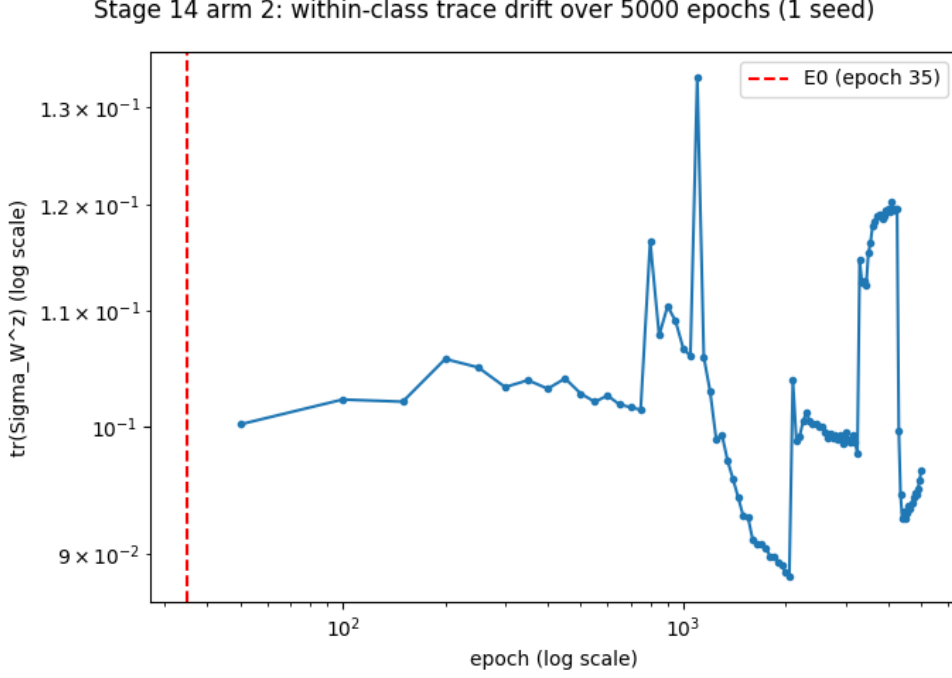


Fig. 6 Within-class z -space trace $\text{tr}(\Sigma_W^z)$ vs. log-epoch over a 5000-epoch horizon, one seed, post-TPT.

5.6 Collapse by rotation, verified at machine precision

We verify Proposition 3 to machine precision. Across five reupload-disabled seeds at 4 qubits, the maximum relative change in total variance across every checkpointed epoch is on the order of 10^{-15} – nine orders of magnitude below our preregistered 10^{-6} threshold – while the same runs’ TPT fraction is 5/5 and NC1_z still decays by a factor of 0.0209. Per-class purity and pairwise class-mean overlap (Corollary 4) are likewise exactly invariant under training in this setting (again on the order of 10^{-15}), a direct structural statement about Du et al.’s optimum: their orthogonal-means optimum is training-unreachable in this architecture unless already present at initialization. Data re-uploading breaks the argument outright: deeper re-uploading circuits correspondingly contract total variance more with depth ($\rho = -0.451$, $p = 0.0035$), and even at the $L=8$ anchor, `reupload=true` `total_var` declines by a small but nonzero 0.73%. Collapse under `reupload=false` is, precisely, not shrinkage: it is training driving $U(\theta)$ to rotate a fixed constellation of states as a rigid body in Hilbert space. Figure 7 overlays the two encoding regimes directly.

5.7 Replication across qubit count: $n = 4, 6, 8$

We replicated the full witness set at 6 and 8 qubits, same dataset recipe lifted to $\mathbb{R}^6$ and $\mathbb{R}^8$. At $n = 6$, the machine-precision total-variance invariance replicates exactly

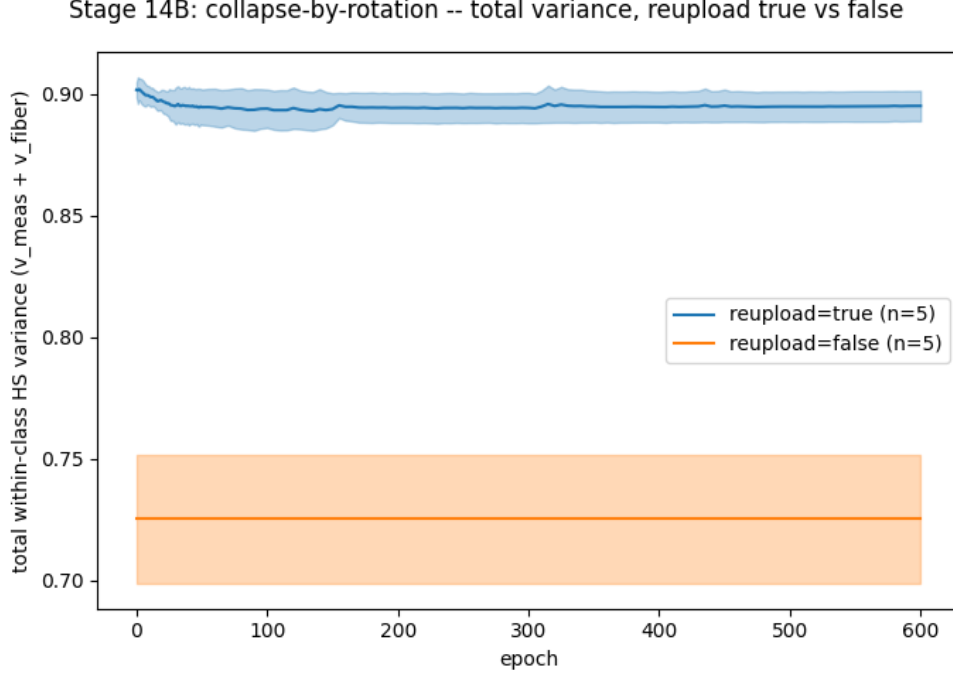


Fig. 7 Total within-class Hilbert–Schmidt variance vs. epoch, data re-uploading enabled vs. disabled, mean $\pm$ 1 std over 5 seeds.

(relative change $\sim 10^{-15}$, 5/5 seeds), as does the NC1_z decay (ratios 0.0028–0.0043), and the fiber fraction strengthens with dimension exactly as Proposition 6 predicts: 0.99963/0.99917 at 6 qubits (reupload true/false) against 0.9934 at 4 qubits. At $n = 8$, the same invariance holds again (relative change 4.75×10^{-15} to 7.91×10^{-15} , 5/5 seeds, `reupload=false`), and `fiber_fraction` exceeds the $n = 6$ reference in 10/10 runs across both re-upload arms (range 0.999667–0.999980) – a replication at a third point, strengthening the dimension-counting trend of Eq. 5 without yet establishing a formal scaling law. Figure 8 compares $n = 4$ and $n = 6$ directly; Figure 9 (Section 8) overlays all three qubit counts against the isotropic-null prediction.

6 Real Vision-Data Results

A central gap in prior work on this mechanism was that it had only been verified on controlled synthetic data. We close that gap here, on MNIST and Fashion-MNIST (classes $\{0, 1, 2\}$).

6.1 The real-data terminal-phase gap and its closure

Under angle encoding, a samples-per-class (spc) budget of 200 never reaches TPT on either dataset (0/26 VQC runs across both re-upload arms and readout families, at $L = 8/16/32$ -probe depths), while a capacity-matched classical MLP on the identical

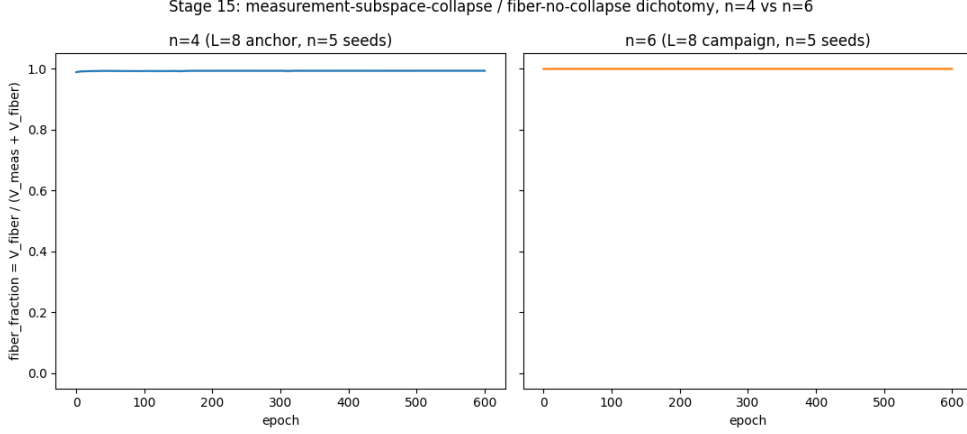


Fig. 8 Fiber fraction vs. epoch, 4-qubit vs. 6-qubit anchors, mean $\pm$ 1 std over 5 seeds each.

Table 1 Real-data TPT attempt, MNIST $\{0, 1, 2\}$, PCA-4, $L=16$, 5 seeds/cell. NC1_z ratio is final/init; total_var is the maximum relative change over training (a range for $\text{reupload}=\text{true}$ ’s mild, seed-varying contraction; a single value for $\text{reupload}=\text{false}$ ’s machine-precision invariance, per Proposition 3).

Variant	spc	TPT	Test acc.	Med. NC1_z	total_var
$\text{reupload}=\text{true}$	50	5/5	0.567	0.00145	0.02–0.04
$\text{reupload}=\text{false}$	50	5/5	0.927	0.02706	$\sim 6 \times 10^{-15}$
$\text{reupload}=\text{true}$	100	5/5	0.703	0.00195	0.02–0.03
$\text{reupload}=\text{false}$	100	4/5	0.940	0.01822	$\sim 7 \times 10^{-15}$

PCA-4 features reaches TPT in 5/6 seeds – a genuine ansatz-expressivity gap on real image data, not a pipeline defect (train accuracy plateaus stably at 0.87–0.96 for hundreds of epochs). Reducing spc to 50–100 closes this gap entirely: Table 1 summarizes both spc cells (MNIST, $L=16$, 5 seeds/cell). NC1_z decays 2–3 orders of magnitude in every TPT-reaching seed, exactly as on synthetic data.

6.2 A clean isolation control: total-variance invariance independent of TPT

The one seed that did *not* reach TPT ($\text{spc}=100$, $\text{reupload}=\text{false}$, one of five) supplies a clean isolation control for Proposition 3: its total_var relative change (5.3×10^{-15}) is indistinguishable from the four TPT-reaching seeds in the same cell, directly demonstrating that the invariance is a structural property of the architecture (fixed encoding, fixed measurement), independent of whether the terminal training regime is ever reached.

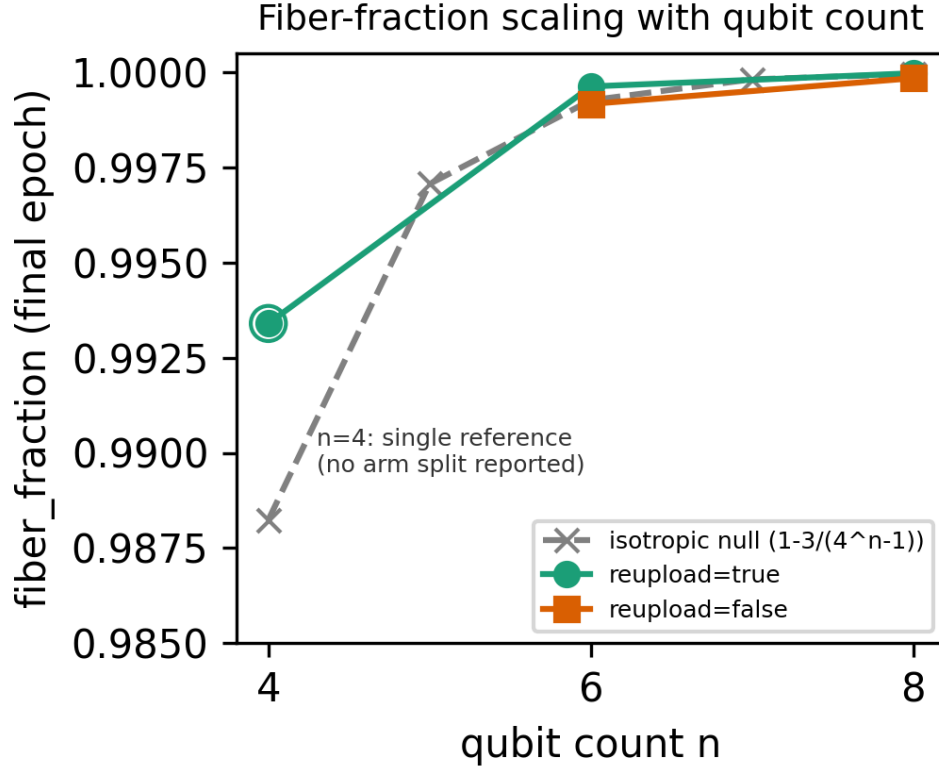


Fig. 9 Fiber fraction vs. qubit count ($n = 4, 6, 8$), synthetic and real-data runs pooled, with the isotropic-null prediction $1 - C/(4^n - 1)$ (Eq. 5) overlaid.

6.3 A second, independent encoding: amplitude embedding

To test whether Proposition 3’s invariance is specific to angle encoding, we repeat the experiment under amplitude embedding (MNIST PCA-16 normalized onto 4 qubits, $\text{spc}=50$, $L=16$). All 4 probed seeds (both re-upload arms) reach TPT, and **reupload=false** total_var is again invariant to machine precision ($4.98\text{--}5.23 \times 10^{-15}$) – the same fingerprint under a structurally different encoding, exactly as Proposition 3 predicts (its hypotheses reference only the encoding being applied once and being parameter-independent, not any specific functional form). Notably, under amplitude encoding **reupload=true** reaches TPT (train accuracy 1.0) but test accuracy sits *at chance* (0.367, $C = 3$), a severe memorization/generalization dissociation co-occurring with a training-side collapse signal (NC1_z ratio ~ 0.003) that looks, in kind, identical to every generalizing arm – the collapse diagnostic alone does not distinguish generalizing from memorizing solutions, and this pattern replicates the reupload-hurts-generalization trend of Section 6.1 under a second encoding family. Figure 10 shows the total-variance trajectory on real data directly.

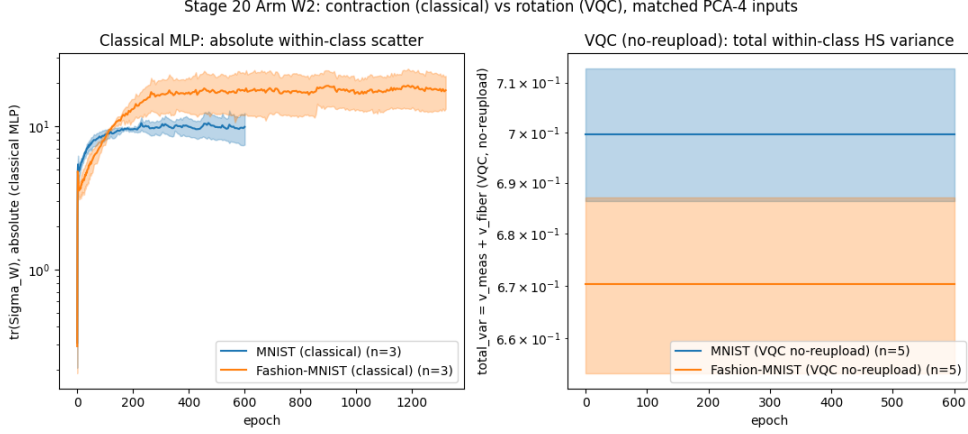


Fig. 10 Left: classical MLP $\text{tr}(\Sigma_W)$ vs. epoch (log scale), both datasets – grows under training. Right: VQC total_var vs. epoch, no-reupload, both datasets – exactly flat.

7 Comparative Study: Classical MLP vs. VQC Variants

7.1 Absolute scale: conservation vs. growth

We compare the quantum representation’s exactly conserved total_var against a matched classical MLP’s analogous absolute within-class scatter, $\text{tr}(\Sigma_W)$ (unnormalized, distinct from the NC1 ratio), on identical PCA-4 features. Across 6 seeds (both datasets), classical $\text{tr}(\Sigma_W)$ *grows* 27–163 $\times$ under training (Figure 10, left) despite weak L2 regularization ($\text{wd} = 5 \times 10^{-4}$, evidently insufficient to bound absolute activation scale), even as the normalized NC1 ratio still falls below 1 (≈ 0.52 mean) – because the between-class scatter Σ_B grows faster still. This does not contradict the feature-space NC1 result of Section 7.2 (ratio ~ 0.515 , feature-space collapse confirmed): NC1 is normalized by Σ_B , and Σ_B grows even faster than Σ_W as ReLU-MLP weights scale up during training. Both representations’ respective collapse ratios fall primarily through *between*-class expansion rather than within-class contraction. The quantum case carries an additional structural guarantee absent classically: Proposition 3’s exact conservation law holds regardless of regularization strength, architecture width, or training duration, whereas the classical pattern is an empirical consequence of this architecture’s specific optimization dynamics, not a theorem. The VQC’s total_var (matched inputs, no-reupload) remains invariant at $\sim 10^{-15}$ throughout (Figure 10, right) – a sharper illustration of the structural-vs.-merely-empirical distinction than a mild classical contraction would have been.

7.2 Comparative tables, split by dataset

Table 2 and Table 3 report the four-approach comparison (classical MLP; VQC no-reupload; VQC reupload; VQC with a trainable measurement, R_3) independently per dataset, confirming the dichotomy of Section 4 is not an artifact of pooling two datasets

Table 2 MNIST {0, 1, 2}, comparative study.

Approach	Seeds	Test acc.	TPT	z -NC1 ratio	Abs. contraction
Classical MLP	3	0.936	2/3	0.458	$31.2\times$ <i>increase</i>
VQC no-reupload	5	0.937	0/5	0.0483	invariant ($\sim 6.7e-15$)
VQC reupload	5	0.688	0/5	0.0046	2.64% contraction
VQC R_3	3	0.694	0/3	2.351 ($\uparrow$)	2.70% contraction

Table 3 Fashion-MNIST {0, 1, 2}, comparative study.

Approach	Seeds	Test acc.	TPT	z -NC1 ratio	Abs. contraction
Classical MLP	3	0.933	3/3	0.572	$82.5\times$ <i>increase</i>
VQC no-reupload	5	0.920	0/5	0.0724	invariant ($\sim 6.2e-15$)
VQC reupload	5	0.738	0/5	0.0046	2.97% contraction
VQC R_3	3	0.731	0/3	1.483 ($\uparrow$)	4.07% contraction

together, and add an “absolute contraction” column showing the classical arm’s unconstrained growth alongside each VQC arm’s invariant-to-mildly-contracting full-space scale. The R_3 arm is the one condition in this study where the z -space collapse direction reverses (NC1 $_z$ *increases*); this arm did not reach TPT in either dataset (test accuracy ≈ 0.71 , tied with the reupload arm for the lowest of the four), so we report this as an underfitting-confounded observation, not a clean statement about measurement-trainability’s causal effect on collapse – the arm never reached the terminal training regime that every other scored NC1 $_z$ -decay result in this paper is gated on.

Pooled numbers (both datasets, 6/10/10/6 seeds respectively) match: test acc. 0.935/0.928/0.713/0.713, z -space ratios 0.515/0.060/0.0046/1.92 – the per-dataset split does not change any verdict, and shows the dichotomy (unbounded classical growth vs. structurally bounded quantum invariance) holds per-dataset, not only in aggregate. The arm with the strongest z -space collapse (VQC reupload, ratio 0.0046) ties for the worst test accuracy with the one arm showing no z -space collapse at all (VQC R_3 , ratio 1.92, an increase), both markedly below the no-reupload VQC arm (0.928) and the classical MLP (0.935); this is an architectural confound (both share `reupload=true`’s altered effective circuit depth), not a clean “more collapse implies worse accuracy” trend.

8 Generality: Optimizer, Ansatz, Dimension, and Noise

To address whether the rotation mechanism and its collapse phenomenology are specific to our default Adam optimizer, `StronglyEntanglingLayers` ansatz, or exact (noiseless) simulation, we ran four breadth-sprint arms at the 4-qubit `blobs3` anchor (qubit-count generality is reported in Section 5.7).

Optimizer generality (SGD, momentum 0.9). A 5-seed campaign at the tuned learning rate reaches TPT 5/5; total.var machine-precision invariance holds 5/5 (all $< 10^{-6}$, matching the Adam-optimizer threshold) and NC1 $_z$ decays at least

one order of magnitude in 5/5 seeds (max ratio 0.0272). The rotation-invariance mechanism and the collapse phenomenology are not Adam-specific.

Ansatz generality (hardware-efficient ansatz). Under a per-qubit R_Y - R_Z rotation layer with a linear-chain CNOT entangler (2 parameters/qubit/layer, vs. `StronglyEntanglingLayers`’ 3), `reupload=false` total_var machine-precision invariance holds 5/5 seeds (all $< 10^{-6}$, same $\sim 10^{-15}$ order of magnitude as the default ansatz), and `reupload=true` reaches TPT 5/5. The rotation-invariance theorem is a property of the no-reupload encoding structure, not of the entangling-layer ansatz family.

Noisy inference (exact depolarizing simulation, inference-only). Restructured as an exact `default.mixed` density-matrix computation over the trained $n = 4$, $L=8$, R_0 anchor’s 180 samples, at two depolarizing probabilities p : the fitted uniform-contraction factor $\lambda_{\text{sim}}(p)$ decreases monotonically with p (0.8874 at $p=0.001$, 0.5500 at $p=0.005$), qualitatively the same phenomenon as the hardware contraction of Section 9. Angular geometry survives at the lower noise level (equiangle-deviation relative change 3.1% at $p=0.001$, under the 10% threshold) but not at the higher one (15.7% at $p=0.005$, inside the preregistered partial band). Class-mean angular geometry survives mild depolarizing noise but visibly distorts at higher noise, an exact-simulation counterpart to the hardware findings of Section 9 rather than a clean confirmation at every tested noise level.

9 Hardware Validation

We validate the mechanism, inference only (no training on hardware, no tomography), against the strictly limited claim it licenses: whether z -space collapse witnesses computed from hardware measurements agree with simulation within shot noise, nothing stronger. We loaded the trained 4-qubit, $L=8$ anchor’s parameters, translated the circuit to a gate-level representation, verified the translation against the exact simulator to better than 10^{-6} on all samples before any hardware submission, and executed on two independent IBM backends at 2048 shots per circuit.

9.1 Primary backend: `ibm_marrakesh`

Hardware NC1_z (0.6749) does not agree with the exact simulator’s value (0.5472) within shot noise: only 17.6% of (sample, class) z -components fall within one shot-noise standard deviation of the simulator, with a mean absolute deviation about $3.4\times$ the mean shot-noise scale. We do not claim agreement; we report the disagreement’s structure. A single scalar contraction fit ($z_{\text{hw}} = \lambda z_{\text{sim}}$) recovers $\lambda = 0.7033$ and $R^2 = 0.9305$, both within our preregistered ranges, while a composite formula predicting NC1_z from that same fit fails catastrophically (predicted 758.7 against an observed 0.6749). We diagnose this as the numerical instability of Remark 2: for any 3-class z -space, the between-class covariance is exactly rank-deficient by construction, and a generic empirical residual does not share its null direction, so the pseudoinverse in NC1_z ’s definition divides by a small, non-noise eigenvalue. Separately, the angular geometry of the class-mean triangle survives the contraction: the exact simulator’s ETF-deviation value falls inside a 1000-resample hardware bootstrap’s 95% confidence

Table 4 Hardware validation, two independent IBM backends, same trained $n = 4$, $L = 8$, R_0 anchor.

Quantity	ibm_marrakesh (HW1-3)	ibm_kingston (B5)
Contraction λ	0.7033	0.6709
Fit R^2	0.9305	0.9356
Simulator equiangle_dev _z	0.0969	0.09691
Hardware bootstrap 95% CI	[0.0831, 0.1152]	[0.0813, 0.1167]
Untrained/trained NC1 _z ratio	94.2×	–

interval (0.0969 inside [0.0831, 0.1152]). An untrained-parameter control on the same circuit gives NC1_z 94.2× the trained hardware value, confirming the hardware pipeline measures a genuine trained-versus-untrained effect rather than a fixed noise floor.

9.2 Second backend: ibm_kingston

To test whether the contraction and angle-survival results are specific to one device, we repeated the identical protocol (same trained anchor, same 180 samples, 2048 shots) on **ibm_kingston**. Table 4 compares both backends directly: the fitted contraction $\lambda_{\text{kingston}} = 0.6709$ falls inside the preregistered band and close to **ibm_marrakesh**’s 0.7033 (backend-consistent), $R^2 = 0.9356$ is a solid linear fit, and the simulator’s equiangle-deviation value (0.09691) again falls inside the hardware bootstrap’s 95% CI ([0.0813, 0.1167]). Both bullets of the preregistered rule hold. As on the first backend, the composite NC1_z-prediction formula (not itself part of the scoring criteria) produces an unstable value on this job (predicted 146.4 vs. observed 0.562, 259× relative error) – a second, independent instance of Remark 2’s diagnosed instability, reported factually rather than discarded.

Figure 11 shows the rescaled **ibm_marrakesh** hardware values against the simulator; Figures B2–B4 (Appendix B) give further hardware and normalized-witness detail.

10 Discussion and Limitations

We have shown that measurement-subspace collapse reconciles a persistent full-Hilbert-space null with real, large collapse in the projection the loss can see, and that under fixed encoding (applied once) and fixed measurement this collapse is exactly a rotation, not a contraction, verified to machine precision on three independent quantities. This replicates at three qubit counts, on real vision data under two independent encodings, under a second optimizer and a second ansatz family, and, within a strictly limited scope, on two independent hardware backends.

Several limitations bound these claims. The dimension-counting trend (Section 5.7) is established at three points ($n = 4, 6, 8$), not as a formal scaling law. Real-data verification required reducing samples-per-class substantially below what the classical baseline needs to interpolate the same features, pointing to a genuine ansatz-expressivity gap on real image data, not a pipeline defect; the R_3 (trainable-measurement) arm’s z-space reversal is confounded by underfitting (0/6 seeds reaching TPT) and does not disentangle whether trainable measurement layers actively work

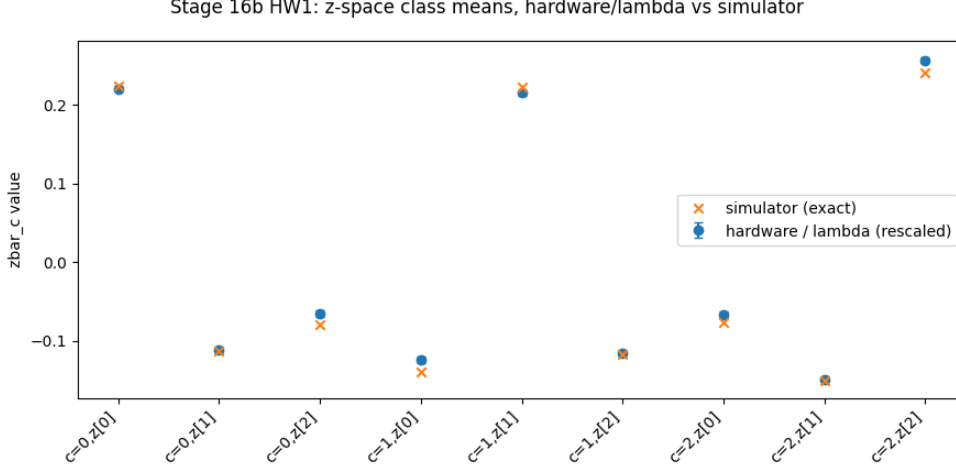


Fig. 11 Hardware class-mean z -values (`ibm_marrakesh`), rescaled by the fitted contraction λ , overlaid on the exact simulator’s values.

against z -space collapse or simply cannot fit these classes well enough to reach the collapse-observing regime. Entanglement entropy is measured at a single fixed bipartition throughout and is explicitly not covered by the rotation invariance (Remark 1). The `reupload=true` full-space contraction on real data (2.0–3.6%, mean 2.8%) is 3–4 $\times$ larger than on synthetic `blobs3` (0.73%), an open quantitative gap not explained here. All hardware evidence is inference-only, two backends, one trained seed per backend plus one untrained control; no repetition across trained seeds on hardware, and no on-hardware training. The dynamical-Lie-algebra capacity bound compared against throughout is an upper bound on the critical parameter count, not the critical parameter count itself.

11 Conclusion

For a broad and practically relevant VQC architecture class, quantum neural collapse under gradient-based training is a rotation of the full representation’s Hilbert–Schmidt geometry, not a contraction – a structural, provable invariance verified to machine precision on a controlled synthetic benchmark, at three qubit counts, under a second optimizer and ansatz family, and, for the first time in this line of work, on real vision data under two independent encoding families. This reframes what “collapse” means for a quantum classifier’s learned representation: the measurement-visible projection collapses because the loss can only ever act there (Lemma 1), while the representation itself is being rigidly reoriented, not shrunk. A matched classical baseline shows the opposite absolute-scale behavior – unconstrained growth rather than conservation – underscoring that the quantum invariance is structural, not an artifact of weak regularization. We report every preregistered prediction made across this campaign’s ten result-bearing stages against its scored outcome (Appendix A), confirmed, refuted, and partial alike.

Declarations

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Ethics approval and consent to participate. Not applicable – this study involves no human participants, human data, or animal subjects.

Consent for publication. Not applicable.

Data Availability. All code, configuration files, preregistered predictions and their scoring thresholds, per-epoch training logs, and figures underlying this paper are available in the project repository: <https://github.com/G26karthik/QNC>.

Materials Availability. Not applicable.

Code Availability. See Data Availability; all analysis code is version-controlled in the same repository.

Author Contribution. G Karthik Koundinya conceived the research questions, designed the mathematical specification and every preregistered prediction with its scoring threshold, reviewed all reported numbers against their source logs, verified the theorem statements and proofs of Section 4, and wrote and is responsible for the final manuscript. Implementation and drafting assistance from an AI coding agent is disclosed in Section 3.4.

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Appendix A Full preregistration ledger

Counting convention. We count each independently-verdicted *leg* of a pre-registered item as one item: if a stage’s findings assign separate CONFIRMED/REFUTED/PARTIAL verdicts to distinct sub-claims under one preregistration header (e.g. a full-space leg and a z -space leg scored in opposite directions), each leg counts separately; a header carrying only one verdict word counts once regardless of how many sub-bullets discuss it narratively. This convention is applied uniformly across every stage, not invented to hit a target number. Items that are pure deliverables rather than directional predictions (a statistics-methodology upgrade, a mechanism-schematic figure, a scaling-figure assembly, or a ledger recount itself) are reported separately below and excluded from the headline tally, consistent with how they were originally registered.

Resolving the internal discrepancy. An earlier internal draft (*paper-ieee*) reported **35** preregistered items (23 CONFIRMED, 6 REFUTED, 5 PARTIAL, 1 factual) – this traces exactly to Stages 12–17b’s 30-item tally plus Stage 18’s B1/B2/B3/B5 (+4 CONFIRMED) and B4 (+1 PARTIAL), but silently drops Stage 18’s B6/B7 and never folds in Stage 19 at all. A separate internal record (*RESULTS_V3.md*) states its own tally as **44** items (27/7/5/5), self-consistent but computed with one-row-per-header counting rather than the leg-split convention, and omitting the scaling-figure deliverable from its own table. Applying the leg-split convention uniformly across Stages 12–19 gives **48** items (29 CONFIRMED, 7 REFUTED, 5 PARTIAL, 7 UNSCOREABLE/factual); excluding the three pure-deliverable items in that range (Stage 18’s B6, B7; Stage 19’s V3) gives **45** (29/7/5/4). Extending the identical leg-split convention through this paper’s own Stage 20 arms (W1–W4, Table A1’s final eight rows) adds 5 CONFIRMED, 1 REFUTED, and 2 UNSCOREABLE/factual items (one of which, W3, is itself the ledger-recount deliverable). The paper’s adopted headline, across all ten stages and excluding the four pure-deliverable items (B6, B7, V3, W3): **52 preregistered items (34 CONFIRMED, 8 REFUTED, 5 PARTIAL, 5 UNSCOREABLE/factual)**; including the four deliverables as their own UNSCOREABLE/factual rows gives **56 (34/8/5/9)**. No verdict is softened or omitted anywhere in Table A1.

Table A1: Every preregistered item across Stages 12–20 and its scored verdict, leg-split convention (see text). Deliverable items (not directional predictions) are marked accordingly.

ID	Stage	Verdict	Prediction (short)	Key number
P1	12	CONFIRMED	$NC1_z/r_z$ decay post-TPT	ratio 0.0045 (L8), 0.0071 (C=2)
P2	12	CONFIRMED	β -mean margin grows post-TPT	$-0.87 \rightarrow 8.55$
P3	12	REFUTED	box vertices beat simplex ETF (C=3)	$d_{\text{boxn}} 0.712$ vs. $d_{\text{etf}} 0.072$; angle alone doesn’t discriminate (Prop. 5)
P4	12	PARTIAL	fiber decomposition dichotomy	fiber_fraction 0.9934 (≥ 0.9 met); V_{fiber} not flat vs. null
P5	12	CONFIRMED	untrained null shows no z -space collapse	$p = 0.00067$

ID	Stage	Verdict	Prediction (short)	Key number
H1	13B	PARTIAL	loss-visible subspace concentrates collapse	concentration $1.42\text{--}1.68\times$ ($<2\times$ threshold)
H2	13B	PARTIAL	fixing depth removes full-space gap	full-space $p \geq 0.42$ (met); fiber-excess $p=0.0079$ (not met)
H3	13B	CONFIRMED	logit NC1_z invariant across families/depths	all pairs $p \geq 0.31$
Arm1	14	CONFIRMED	$\beta_{\max}=500$ changes nothing	bit-identical to baseline, $p=1$
Arm2	14	CONFIRMED	no slow drift over 5000 epochs	$\rho = -0.072$, $p=0.48$
Arm3	14	REFUTED	label smoothing accelerates shrink	right direction, $p=0.42$
Arm4	14	CONFIRMED	TPT fraction non-decreasing, $L \in \{3, 5, 6\}$	$0.3 \rightarrow 1.0 \rightarrow 1.0$
Arm5	14	REFUTED	single-input QFI rank saturates near 255	saturates at 30 for all L
Arm6	14	REFUTED	frozen β forces within-class collapse	wrong direction, $p=0.016$; E_0 delayed $\sim 4.1\times$
HR1	14B	CONFIRMED	reupload=true total_var decline $<10\%$	0.73% decline
HR2	14B	CONFIRMED	reupload=false total_var exactly constant	rel. change $\sim 10^{-15}$, 5/5 seeds
HR3	14B	CONFIRMED	reupload=false NC1_z still decays ≥ 1 order	ratio 0.0209, 5/5 TPT
HR4	14B	CONFIRMED	deeper reuploading circuits contract more	$\rho = -0.451$, $p=0.0035$
HR5	14B	REFUTED	variance floor scales as blob_std^2	power 0.119 (outside [1,3])
HR5b	15	CONFIRMED	epoch-0 spread flat across blob_std	rel. range 2.79%/6.67% ($<20\%$)
HR5c	15	PARTIAL	final trace_W correlates with own epoch-0	$\rho=0.132>0$, $p=0.53$ (n.s.)
HN1	15	CONFIRMED	reupload=false total_var constant at $n=6$	rel. change $\sim 5\times 10^{-15}$, 5/5 seeds
HN2	15	CONFIRMED	fiber_fraction at $n=6$ exceeds $n=4$'s 0.993	0.99963/0.99917
HN3	15	CONFIRMED	NC1_z decays ≥ 1 order at $n=6$	ratios 0.0028/0.0043
HN4	15	UNSCOREABLE	TPT fractions (factual)	8/8 depth-probe cells, campaign 5/5 both arms
HW1	16b	REFUTED	uniform-contraction fit accounts for NC1_z rise	$\lambda=0.7033$, $R^2=0.9305$ met; NC1_z -pred. off by $\sim 1123\times$
HW2	16b	CONFIRMED	z -space angles invariant under contraction	sim. 0.0969 inside hw. 95% CI [0.0831, 0.1152]
HW3	16b	CONFIRMED	untrained hardware $\text{NC1}_z \geq 10\times$ trained	ratio 94.2%
HR6	17b	CONFIRMED	purity + pairwise overlap constant, $n=4$	rel. change $\sim 10^{-15}$, both, 5/5 seeds
HN5	17b	CONFIRMED	purity + pairwise overlap constant, $n=6$	rel. change $\sim 10^{-15}$, both, 5/5 seeds
B1	18	CONFIRMED	SGD momentum optimizer generality	TPT 5/5, total_var and NC1_z replicate
B2	18	CONFIRMED	hardware-efficient ansatz generality	total_var invariant 5/5, TPT 5/5 (reupload=true)
B3	18	CONFIRMED	$n = 8$ replication	total_var invariant 5/5; fiber.fraction $> n=6$ in 10/10
B4	18	PARTIAL	noisy simulation preserves angle structure	holds at $p = 0.001$, not at $p = 0.005$
B5	18	CONFIRMED	second hardware backend (ibm_kingston)	$\lambda=0.6709$, $R^2=0.9356$; angle-survival holds
B6	18	(deliverable)	statistics upgrade	done
B7	18	(deliverable)	mechanism schematic	done
V1(i)	19	CONFIRMED	reupload=false total_var invariant, real data	rel. change $4.5\text{e-}15\text{--}9.5\text{e-}15$, 5/5, both datasets
V1(ii)	19	UNSCOREABLE	NC1_z decay ≥ 1 order, TPT-reaching seeds	0/20 seeds reached TPT – precondition unmet
V1(iii)	19	UNSCOREABLE	TPT fractions (factual)	0/26 VQC runs reached TPT at $L=8/16/32$ -probe
V2(a)-feat	19	CONFIRMED	classical MLP collapses feature space	NC1 ratio 0.31–0.91, 5/6 TPT
V2(a)-full	19	UNSCOREABLE	classical MLP collapses full input space	not well-posed for a fixed (untrained) input
V2(b)	19	CONFIRMED	VQC no-reupload: z -space collapse, full-space invariant	full-space $\sim 1\text{e-}14\text{--}1\text{e-}15$; z -ratio 0.060 mean
V2(c)-z	19	CONFIRMED	VQC reupload: z -space collapse	ratio 0.0046 mean

ID	Stage	Verdict	Prediction (short)	Key number
V2(c)-full	19	CONFIRMED	VQC reupload: small full-space contraction	2.8% mean
V2(d)-z	19	REFUTED	VQC R_3 : z -space collapse	NC1 _z <i>increased</i> , ratio 1.92 mean, 5/6 seeds
V2(d)-full	19	CONFIRMED	VQC R_3 : small full-space contraction	3.4% mean
V3	19	(deliverable)	fiber-fraction scaling figure	done, Figure 9
W1(i)	20	CONFIRMED	TPT fraction, $\text{spc}=50 \geq \text{spc}=100$	10/10 vs. 9/10
W1(ii)	20	CONFIRMED	NC1 _z decay in TPT-reaching seeds	median ratio <0.1 , both cells, both arms
W1(iii)	20	CONFIRMED	reupload=false total_var invariance	$\sim 10^{-15}$, all 10 seeds incl. non-TPT seed
W2-classical	20	REFUTED	classical $\text{tr}(\Sigma_W)$ contracts <0.5	GROWS 27–163 $\times$ (wrong direction)
W2-VQC	20	CONFIRMED	VQC total_var invariance (matched inputs)	$\sim 10^{-15}$, 10/10 seeds
W3	20	(deliverable)	ledger recount + table split	done (this appendix)
W4-TPT	20	UNSCOREABLE	amplitude-encoding TPT (factual probe)	4/4 TPT, both re-upload arms
W4-var	20	CONFIRMED	total_var invariance, second encoding	$4.98\text{--}5.23 \times 10^{-15}$

Appendix B Additional figures and traceability

Figure B1 shows the entanglement-ablation capacity bound directly; Figure B2 shows the normalized-witness reanalysis confirming the full-space null of Section 5.1 is not a metric artifact; Figure B3 decomposes NC1_z into its within-class and between-class traces; Figure B4 compares hardware and simulator NC1_z directly as a bar chart. Full run-identifier traceability for every figure and table in this paper is recorded in the underlying project findings records (per-stage `STAGE*_FINDINGS.md` files and `RESULTS_V3.md`), included in the data-availability repository (Declarations) but not duplicated here.

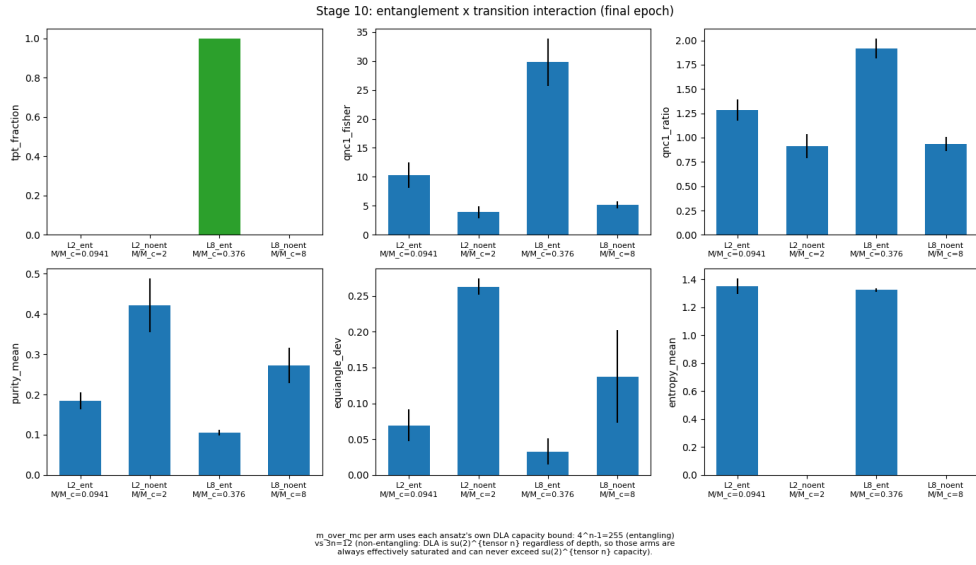


Fig. B1 Entanglement ablation: TPT fraction vs. parameter count, entangling vs. non-entangling ansatz, corrected dynamical-Lie-algebra bounds.

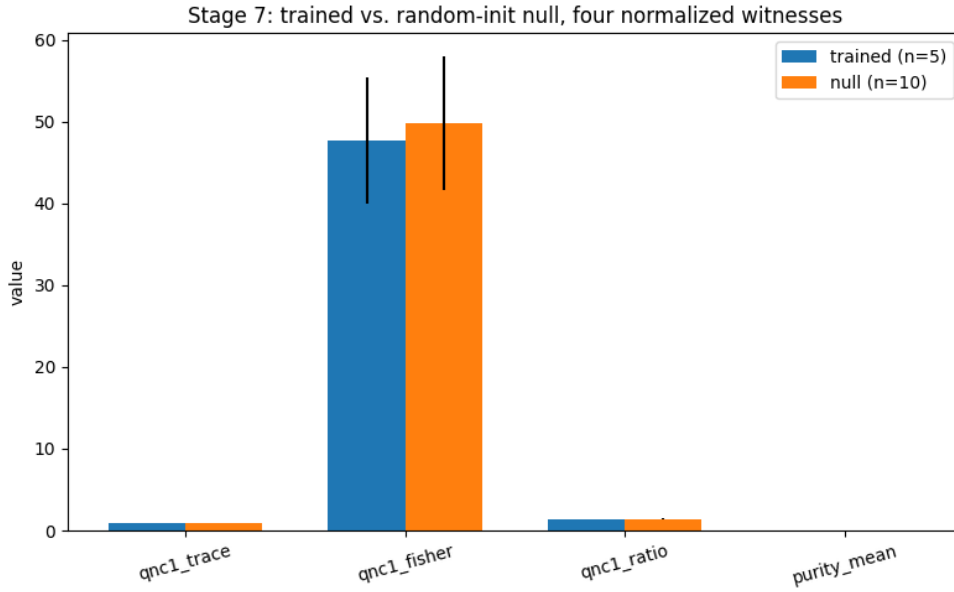


Fig. B2 Four normalized collapse witnesses, trained vs. untrained null, 2-class anchor.

plement: NC1_z decomposition -- within-shrink vs between-growth (mean +/- 1 std,

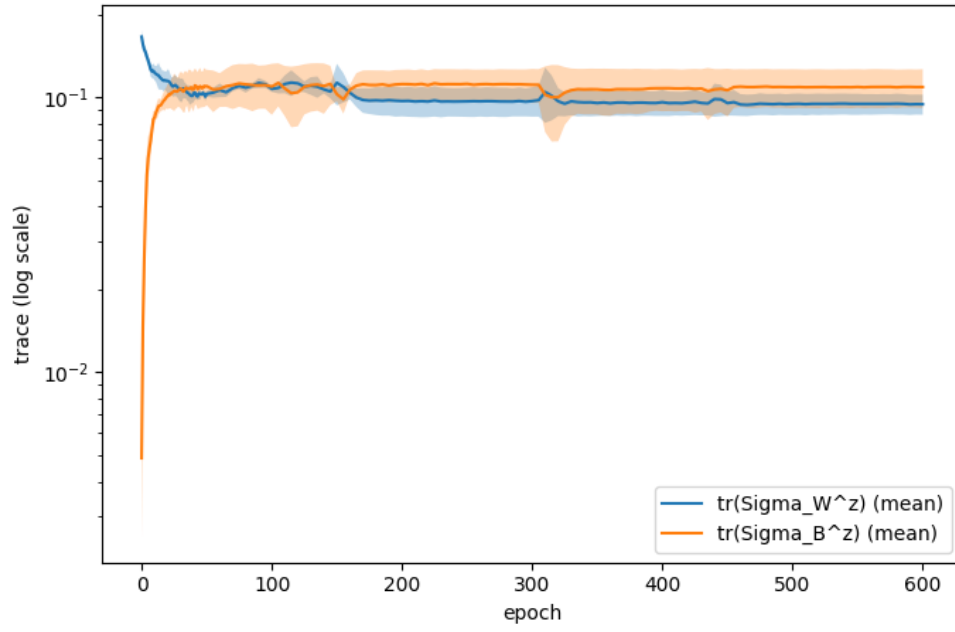


Fig. B3 $\text{tr}(\Sigma_W^z)$ and $\text{tr}(\Sigma_B^z)$ vs. epoch, 4-qubit $L=8$ anchor.

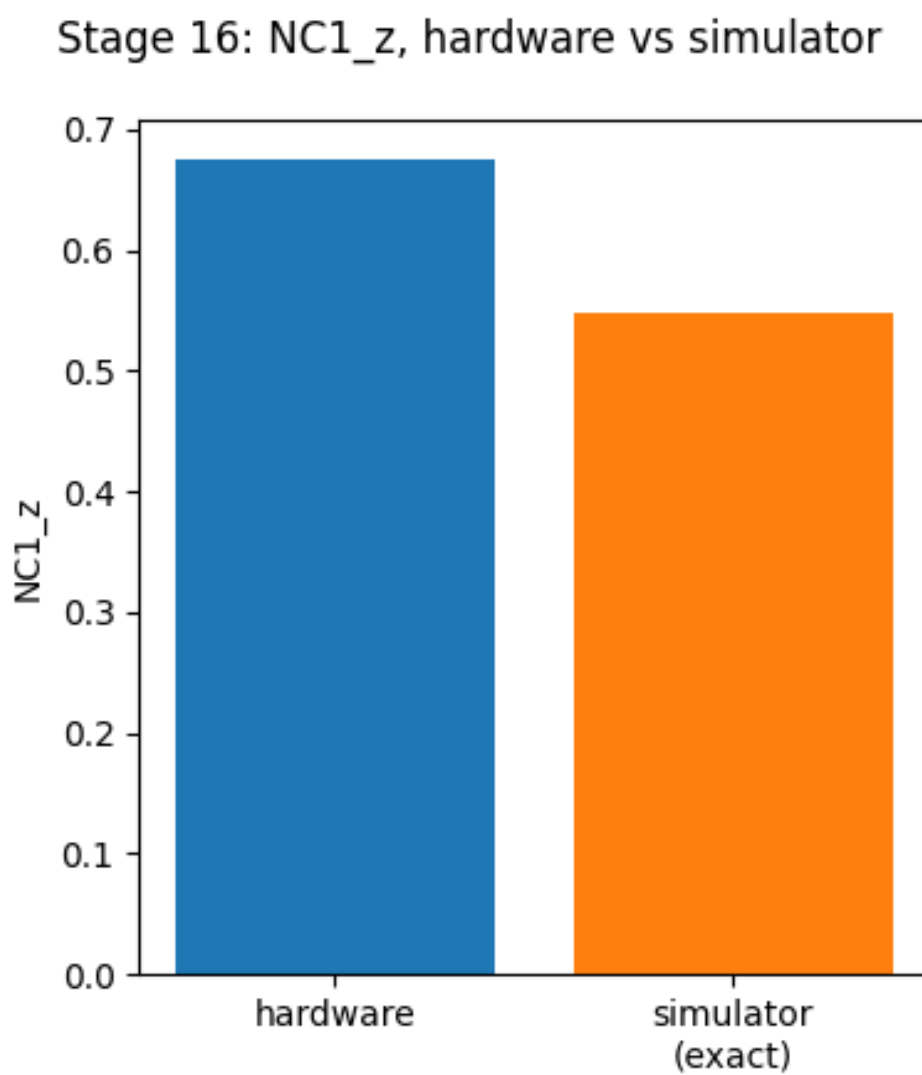


Fig. B4 NC1_z, hardware (ibm_marrakesh) vs. exact simulator, bar comparison.